

Anti-Jamming Trajectory Design for UAV-Enabled Wireless Sensor Networks Using Communication Flight Corridor

Binbin Wu¹, Bangning Zhang¹, Daoxing Guo^{1*}, Hongbin Wang¹, Hao Jiang^{2,3}

¹ College of Communications Engineering, Army Engineering University of PLA, Nanjing 210007, China

² College of Artificial Intelligence, Nanjing University of Information Science and Technology, 210044, China

³ National Mobile Communications Research Laboratory, Southeast University, Nanjing 210096, China

* The corresponding author, email: xyzgfg@sina.com

Abstract: This paper investigates the anti-jamming trajectory design to safeguard the effective data collection, where a unmanned aerial vehicle (UAV) is dispatched to collect data over multiple sensor nodes(SNs) in jamming environment. Under the limited power and transmission range of SNs, we aim to minimize the UAV's flight energy consumption in a finite task period, by jointly optimizing SNs collection sequence and UAV flight trajectory. Firstly, we propose a general optimization framework which consists of path planning and trajectory optimization for the formulated non-convex problem. In the path planning phase, a dynamic programming (DP) algorithm is used to provide the initial path of the UAV, which is the shortest path to visit each SN. In the trajectory optimization phase, we introduce the concept of Communication Flight Corridor (CFC) to meet the non-convex constraints and apply a piecewise Bézier curve, based on Bernoulli polynomial, to represent the flight trajectory of the UAV, which can transform the optimization variables from infinite time variables to polynomial coefficients of finite order. Finally, we simulate the flight trajectory of UAV in hovering mode and continuous flight mode under different parameters, and the simulation results demonstrate the effectiveness of the proposed method.

Keywords: UAV; data collection; jamming environ-

ment; path planning; trajectory optimization

I. INTRODUCTION

1.1 Background and Motivations

Due to the advantages such as fast deployment, flexibility, wide range coverage and line-of-sight (LoS) channels, unmanned aerial vehicle (UAV) communication has attracted extensive attention from both academia and industry. With the rapid commercial use of the fifth-generation mobile networks and the increasing interest on sixth-generation mobile networks, UAV communication has been regarded as an important component of the future integrated space-air-ground network and the strong support for the concept of the internet of everything[1–3].

At the same time, UAV communication also faces several challenges. On the one hand, the broadcasting nature of radio propagation makes the UAV particularly vulnerable to jamming attacks, although the traditional anti-jamming approaches based on the frequency domain, time domain, power domain and space domain methods such as frequency hopping, burst communication, adaptive power transmission and the beamforming technology have been widely used[4, 5]. Thanks to the UAV's highly controllable maneuverability, it is also appealing to defend against the wireless jamming attack in the spatial domain by retreating away from the jammers for the channel quality is dominated by the large scale [6, 7].

Received: Oct. 04, 2021

Revised: Nov. 14, 2021

Editor: Gengxin Zhang

On the other hand, although the trajectory can be regarded as a new optimization dimension, which can be jointly optimized with communication resources and thus brings more system gain, the diverse communication scenarios and quality of service (QoS) requirements greatly increase the difficulty of communication system modeling and optimization of the joint communication-flight framework that considers both the communication resource allocation and flight control[8, 9].

With UAV communication as the research background, this paper aims to realize the collection of sensor node (SN) information that can be correctly demodulated by the UAV or information processing center. The optimization goal is minimizing energy to overcome the problems of limited power and transmission range of SNs under jamming environment by a reliable and efficient UAV trajectory.

1.2 Prior Work

The use of UAV to collect data from distributed nodes on the ground is one of the key technologies of “Internet of Everything”, and there have been a lot of researches. The optimization goals are usually to minimize the mission time of UAV[10–12], minimize flight energy consumption[13–16], maximize the amount of data collection[17, 18], and maximize the number of service sensors[19].

Owing to the SNs can be either located in remote areas with limited wireless coverage or in dense areas with relatively low quality of service, the authors in [19] jointly optimized the trajectory of a UAV and the radio resource allocation to maximize the number of served SNs for time-constrained data collection by applying the branch-and-bound (BRB) method and the successive convex approximation (SCA). Since UAVs are powered by batteries, energy efficient communication is important. Energy minimization for wireless communication with rotary-wing UAV was studied in [20], the authors derived a closed-form propulsion power consumption model for rotary-wing UAV, and then jointly optimized the UAV trajectory and communication allocation among SNs, as well as the total mission completion time by applying SCA technique. The effective transmission range between UAV and SNs was considered in [21], the authors proposed a novel concept of virtual base station (VSB) place-

ment to ensure a desired high communication probability and minimized the mission completion time by applying convex optimization.

In addition, anti-jamming trajectories were designed in [22–24] to ensure the reliable communication between UAV and SNs by taking advantage of the flexible mobility of UAVs. The authors in [25, 26] maximized the minimum (average) expected rate among SNs over a finite flight period, by jointly optimizing the SNs transmission scheduling, the UAV horizontal and vertical trajectory. According to the UAV data collection scenario in the jamming environment in [26], we further consider the communication link constraint of the sensors which completed the data collection mission. The problem of lifetime maximization of a UAV-assisted network in the presence of multiple sources of jammer was investigated in [7], where the UAVs are deployed to collect data from a set of wireless SNs. In [27], anti-jamming strategies based on reinforcement learning were investigated to improve the static UAV-enabled communications against smart jamming attacks.

All the above mentioned studies concentrate on the ground-to-air (G2A) communications. The G2A communication is a key part in UAV-enabled communication systems[28–30]. But, there are two shortcomings in the past research. Most studies depended on line-of-sight (LoS) probabilities and non-line-of-sight (NLoS) probabilities, and discretized the time horizon T into M equally spaced time slots, i.e., $T = M\delta_t$, with δ_t denoting the elemental slot length. The elemental slot is appropriately chosen for the distance between the UAV and the SNs can be assumed to be approximately constant. However, the SNs in the jungle of a mountain or a built-up area usually will produce reflection, refraction, diffraction phenomena, such as simply assume that LoS or NLoS channel is not reasonable.

Meanwhile, the aforementioned works on UAV trajectory optimization studies are mostly based on small-scale scenes, where SCA, heuristic algorithm and reinforcement learning can be effectively applied and the common feature is that the trajectory and flight time of the UAV need to be discretized. However, the slot numbers will lead to “the dimension explosion” in the optimization problem, when the UAV flight lasts for a long time. Furthermore, in the jamming environment, the past studies did not consider whether the collected data could be correctly demodulated, and

it is not reliable to simply meet the demand of data collection amount. The authors in [31–35] propose a method to generate a dynamically feasible trajectory for UAV using the concept of a Safe Flight Corridor (SFC) in an obstacle-cluttered environment. They also introduce a novel research framework, which consists of path planning and trajectory optimization.

Inspired by their work in [31–33], we propose a Communication Flight Corridor (CFC) for UAV's trajectory generation in the jamming environment, which can ensure reliable communication between UAV and SN. Furthermore, we use the Bézier curve to replace the normal polynomial curve in [32], which can significantly reduce the computational complexity. What's more, the Bézier curve can represent the flight trajectory of UAV in continuous time domain, which can avoid the problem of dimensional explosion caused by the large number of discrete time, and the complexity of optimization algorithm can be greatly reduced by optimizing the finite number of coefficients of polynomial. Different from [32], in our UAV-enabled data collection scenario, on the one hand, we need to consider the number and intensity of interference sources around SN when constructing CFC, on the other hand, we also need to ensure the collection time of each SN.

1.3 Contributions and Organization

This paper investigates the anti-jamming optimization problem of UAV performing data collection missions in a jamming environment. The flight trajectory of the UAV is designed and optimized, where the sensor transmits data within the fixed minimum communication rate when the communication ability of the low-power SNs are considered. The main contributions of this paper are as follows:

- We formulate an general optimization framework which consist of path planning and trajectory optimization for minimizing the energy consumption of UAV data collection under the jamming environment while ensuring the constraints, including a maximum speed and accelerated speed of UAV, the minimum data requirement for each SN. This framework gives the best path to visit SNs, which guides the trajectory optimization of UAV. Unlike prior studies, this work can decompose the complex non-convex optimization problem into two steps and reduce the complexity of the original

problem by solving the subproblems.

- To the best of our knowledge, we first put forward the concept of CFC, which can ensure that the data collected by the UAV can be correctly demodulated in the jamming environment. According to the jamming intensity detected by UAV and the known SN position, the CFC is obtained by geometric method. Compared with the conventional methods, the optimization problem, constructed by CFC, is the standard quadratically constrained quadratic program (QCQP), which has low complexity and can be solved effectively.
- We also propose a piecewise trajectory of UAV based on the representation of Bézier curve to transform the infinite time variables into the finite polynomial coefficient variables. And the boundary constraints, motion constraints, continuity constraints and CFC constraints in the optimization problem can be more concise due to the properties of Bézier curve.

The rest of this paper is organized as follows. In section II, We illustrate the investigatory scenario and formulate the optimization problem. Then, we design the algorithm framework and CFC in section III. The trajectory constraint and representation in the trajectory optimization problem are studied and analyzed in section IV. Simulation results are presented to verify the effectiveness of the proposed algorithms in section V. Finally, we conclude this paper in section VI.

Notations: In this paper, The boldface letters denote matrixs. For a vector a , $\|a\|$ represents its Euclidean norm and $a^\dagger$ denotes its transpose. For time-dependent function $x(t)$, $\dot{x}(t)$ denotes the derivative with respect to time t . $\mathcal{W} \times \mathcal{A}$ denotes the cartesian product of set $\mathcal{W}$ and $\mathcal{A}$.

II. SYSTEM MODEL

2.1 System Model

In this scenario, we dispatch a UAV to fly from the starting point to the endpoint to complete the data collection mission of K SNs within T time. The trajectory of the UAV at time t is expressed as $w(t) = (x(t), y(t), H)$, $0 \leq t \leq T$, where H is the fixed altitude. Refer to [21], we set the minimum flight height of the UAV to be greater than the height of the obstacles in the mission area.

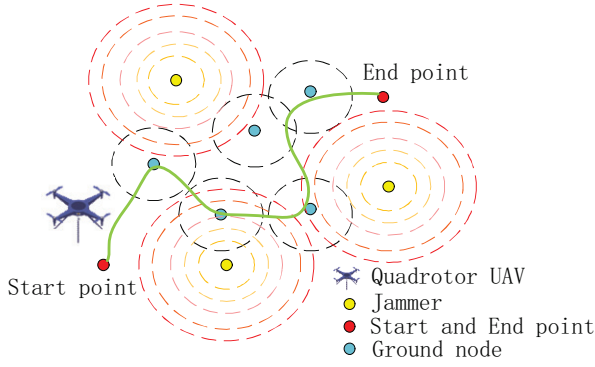


Figure 1. UAV-enabled data collection under jamming environments.

As shown in Figure 1, we consider K SNs in this wireless communication system, represented as a set $\mathcal{K} = \{1, 2, \dots, K\}$, and the position of k -th SN is $w_k = (x_k, y_k, 0)$, $k \in \mathcal{K}$. In addition, there are M jammers in the environment, which are represented as set $\mathcal{M} = \{1, 2, \dots, M\}$, and the position of the m -th jammers is $w_m = (x_m, y_m, 0)$, $m \in \mathcal{M}$. We assume that the position of SNs and jammers are fixed on the ground, UAVs could be get through some standard positioning technology, such as the beidou system, etc. [36]. Combining with the radiation source detection technology, the real-time intensity of jamming can be obtained [37]. Dashed lines of different colors in Figure 1 form outward radiating circles representing the intensity contour lines of the jammers.

2.2 Channel Model

In this paper, we focus on the trajectory optimization of UAV. The channel gain between the SN k and the UAV is represented by [21]

$$h_k(t) = \sqrt{\mu_k(t)}, \quad (1)$$

where $\mu_k(t)$ is the large-scale fading effects at time t , which are affected by the uncertain external environment. $\mu_k(t)$ depends on the line-of-sight (LoS) probabilities and non-line-of-sight (NLoS) probabilities, which can be modeled as

$$\mu_k(t) = \begin{cases} \mu_0 d_k^{-\alpha}(t), & \text{LoS link} \\ a_0 \mu_0 d_k^{-\alpha}(t), & \text{NLoS link} \end{cases}, \quad (2)$$

where $d_k(t) = \|w(t) - w_k\|$ denotes the distance between the SN k and UAV, μ_0 denotes the path loss at the reference distance, $\alpha \geq 2$ is path loss exponent, and $a_0 < 1$ is the attenuation factor under the NLoS link. $P_{k,LoS}(t)$ denotes the probability of LoS between SN k and UAV at time t , which can be calculated approximately by

$$P_{k,LoS}(t) = \frac{1}{1 + b \exp(-c(\beta_k(t) - b))}, \quad (3)$$

where b and c are parameters determined by the external environment, and $\beta_k(t) = \frac{180}{\pi} \sin^{-1}(\frac{H}{d_k(t)})$ is the elevation angle in degree. The average channel gain between SN k and the UAV at time t is expressed as

$$E[|h_k(t)|^2] = P_{k,LoS}(t) \mu_0 d_k^{-\alpha}(t) + P_{k,NLoS}(t) a_0 \mu_0 d_k^{-\alpha}(t), \quad (4)$$

where $P_{k,NLoS}(t) = 1 - P_{k,LoS}(t)$ is the probability of NLoS between SN k and the UAV at time t .

The transmission capacity between the UAV and sensor K at time t , which is given as

$$R_k(t) = B \cdot \log_2 \left(1 + \frac{p_k g_k(t)}{\sum_{m=1}^M p_m g_m(t) + \sigma^2} \right), \quad (5)$$

where $p_k, k \in \mathcal{K}$ is the transmission power of the k -th SN, $p_m, m \in \mathcal{M}$ is the transmission power of the m -th jammer, B is the total channel bandwidth in hertz (Hz), σ^2 denotes the noise power spectral density. According to the statistical characteristics in [21], we approximate the channel gain parameter as $\beta_0 = \bar{P}_{k,LoS} \mu_0 + \bar{P}_{k,NLoS} a_0 \mu_0$. Therefore, we can obtain that the channel gain between the UAV and SN k is $g_k(t) = \beta_0 \|w(t) - w_k\|^{-\alpha}$ and the channel gain between the UAV and the jammer m was $g_m(t) = \beta_0 \|w(t) - w_m\|^{-\alpha}$. The total jamming power of the UAV is $\sum_{m=1}^M p_m g_m(t)$ at the current moment.

2.3 Motion Primitives

Let $s(t) \in \mathcal{S}$ be a dynamical system state, consisting of 2-D position $w(t)$ and its $(n - 1)$ derivatives (velocity, acceleration, jerk, etc.). $\mathcal{S}^{free} \subset \mathcal{S}$ denote free regoin of state space that including UAV

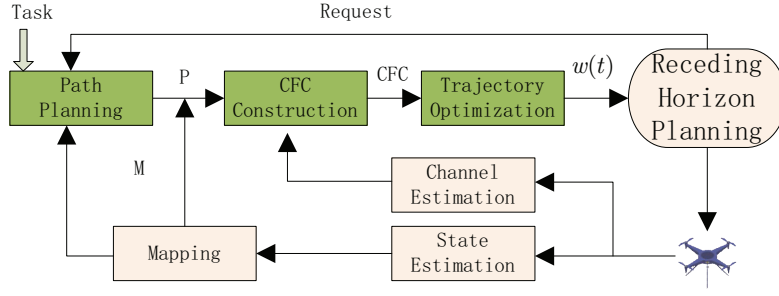


Figure 2. Block diagram of our autonomous system. We first find a valid path in a grid map toward data collection mission, based on which we construct the CFC through convex decomposition. The trajectory inside the CFC is achieved from solving a optimization problem. And in the end we are able to get the desired control commands for navigating the UAV.

flight positions $\mathcal{W}^{free}$, velocity $\mathcal{V}^{free}$ and acceleration $\mathcal{A}^{free}$ on the system's dynamics, such as maximum velocity $v_{\max}$, acceleration $a_{\max}$, and higher order derivatives in each axis. Note that $\mathcal{W}^{free}$ is bounded by size of the map and jammer that we are planning in. Thus, $\mathcal{S}^{free} = \mathcal{W}^{free} \times \mathcal{V}^{free} \times \mathcal{A}^{free} = \mathcal{W}^{free} \times [-v_{\max}, v_{\max}] \times [-a_{\max}, a_{\max}]$.

As described in [32], the differential flatness of UAV systems allow us to construct control inputs from 1-D time parametrized polynomial trajectories specified independently in each pf the two position axes. Thus, we consider polynomial state trajectory $s(t) := [w_D(t)^T, \dot{w}_D(t)^T, \dots, w_D^{(n-1)}(t)^T]^T$, where

$$w_D(t) = \sum_{i=0}^n p_i t^i, \quad (6)$$

and D represents a dimension indicator. To simplify the notation, the system's velocity is denoted as $v(t) = \dot{w}_D(t)^T$, acceleration is $a(t) = \ddot{w}_D(t)^T$, jerk is $j(t) = \dddot{w}_D(t)^T$, etc. Polynomial trajectories of the form Eq. (6) can be generated by considering a linear time-invariant dynamical system $w_D(t) = \mu(t)$, where the control input is $\mu(t) \in \mathcal{U} = [-\mu_{\max}, \mu_{\max}]^2$. In state space form, we obtain a system as

$$\dot{s}(t) = \mathbf{A}s(t) + \mathbf{B}\mu, \quad (7)$$

where

$$\mathbf{A} = \begin{bmatrix} 0 & I_2 & 0 & \cdots & 0 \\ 0 & 0 & I_2 & \cdots & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & \cdots & \cdots & 0 & I_2 \\ 0 & \cdots & \cdots & 0 & 0 \end{bmatrix}, \quad (8)$$

and

$$\mathbf{B} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ I_2 \end{bmatrix}. \quad (9)$$

The resulting trajectory of the linear time-invariant system in Eq. (7) is given as:

$$s(t) = e^{\mathbf{A}t} \mathbf{s}_0 + \int_0^t e^{\mathbf{A}(t-\tau)} \mathbf{B} \mu(\tau) d\tau, \quad (10)$$

where $\mathbf{s}_0 = [w_0, v_0, a_0, \dots]^T$ is the initial state, w_0 is the initial position, v_0 is the initial velocity, and a_0 is the initial acceleration.

2.4 Problem Formulation

The optimization goal is to minimize the energy consumption of the UAVs when performed data collection missions. The UAV energy consumption is in general composed of two main components, namely the communication related energy and the propulsion energy[21]. The communication related energy includes that for communication circuitry, signal pro-

cessing, signal radiation/reception, etc. In this paper, we consider that the communication related power is a constant and collecting data at a fixed communication rate $R_{\min}$ that can be demodulated. On the other hand, the propulsion energy consumption is required to keep the UAV aloft and support its movement. In general, the propulsion energy depends on the UAV flying velocity, acceleration, jerk and so on. In this paper, we are interested in the smoothness or effort of a trajectory as the square L^2 -norm of the control input $\mu(t) = w^{(4)}(t)$, which represent the energy cost in the dynamic system. Therefore, we formulate the optimization problem as follows

$$(P1) \quad \min \int_0^T \|\mu(t)\|^2 dt, \quad (11a)$$

$$s.t. \quad \int_0^T \kappa(t) \cdot R_{\min} dt \geq Q_{\min}^k, \quad k \in \mathcal{K}, \quad (11b)$$

$$\kappa(t) = \mathbb{1}[\max\{R_k(t) - R_{\min}, 0\}]^+, \quad (11c)$$

$$v_{\min} \leq \dot{w}(t) \leq v_{\max}, \quad (11d)$$

$$a_{\min} \leq \ddot{w}(t) \leq a_{\max}, \quad (11e)$$

$$w(0) = w_0, w(T) = w_E, \quad (11f)$$

$$\dot{\mathbf{s}}(t) = \mathbf{A}\mathbf{s}(t) + \mathbf{B}\mu(t), \quad (11g)$$

where $\mathbb{1}[x]^+$ is the indicator function that equals unity if $x > 0$ and zero otherwise. Therefore, constraint condition (11b) is to ensure that the amount of data needed to be collected by SN k within mission time T is not less than $Q_{\min}^k$ and the constraint condition (11c) is to meet the requirements of the low-power transmission system, where $R_{\min}$ is the minimum communication rate that the UAV and SN straight line need to reach. The problem $P1$ is difficult to solve because of the non-convex constraints (11b) and (11c). In the following section, we propose a path planning and trajectory optimization research framework to solve problem $P1$.

III. ALGORITHM FRAMEWORK AND COMMUNICATION FLIGHT CORRIDOR

3.1 Algorithm Framework

In this section, we design the overall structure of the autonomous control system with referencing to [32] for

UAV data collection. We discuss three components through deriving the desired trajectory for controlling the UAV to collect data. The green block diagram in Figure 2 is the content discussed. First, path planning is based on the data collection mission-driven, which defines the problem as a TSP problem from the starting point to the end point. Define the TSP optimization problem as follows

$$(P2) \quad \min \sum_{j \in [1, K]} c_{0,j} \varepsilon_{0,j} + \sum_{i,j \in E} c_{ij} \varepsilon_{ij} + \sum_{i \in [1, K]} c_{i,K+1} \varepsilon_{i,K+1}, \quad (12a)$$

$$s.t. \quad \sum_{j \in [1, K]} \varepsilon_{i,j} = 1, \quad (12b)$$

$$\sum_{i \in [1, K]} \varepsilon_{i,j} = 1, \quad (12c)$$

$$\sum_{j \in [1, K]} \varepsilon_{0,j} = 1, \quad (12d)$$

$$\sum_{i \in [1, K]} \varepsilon_{i,K+1} = 1, \quad (12e)$$

$$\sum_{j \in [1, K]} \sum_{i \in [1, K]} \varepsilon_{i,j} = K, \quad (12f)$$

$$\varepsilon_{i,j} \in \{0, 1\}. \quad (12g)$$

where $i, j \in E = \{[1, K]^2\}$ represents the edge of each SN connection. ε_{ij} indicates whether the next node of i is j . The overhead between each node is denoted by c_{ij} . Constraints (12b) and (12c) indicate the number of passed nodes. Constraints (12d) means the UAV must start from the starting point, and (12e) indicates that the final node must be the end point. The optimization problem is that $P2$ is a TSP, which traverses each intermediate node from the starting point to the end point. There are many mature TSP solving algorithms, such as dynamic programming (DP) [38], simulated annealing algorithm [39], genetic algorithm [40], branch and bound method [41] etc.. In this paper, we adopt a DP algorithm with a global polynomial solution to solve the initial path of the UAV data collection mission, and the detailed procedure of the algorithm is given in Appendix A.

3.2 Communication Flight Corridor

Based on the initial path, the appropriate waypoints are selected to divide the trajectory [32]. The trajectory

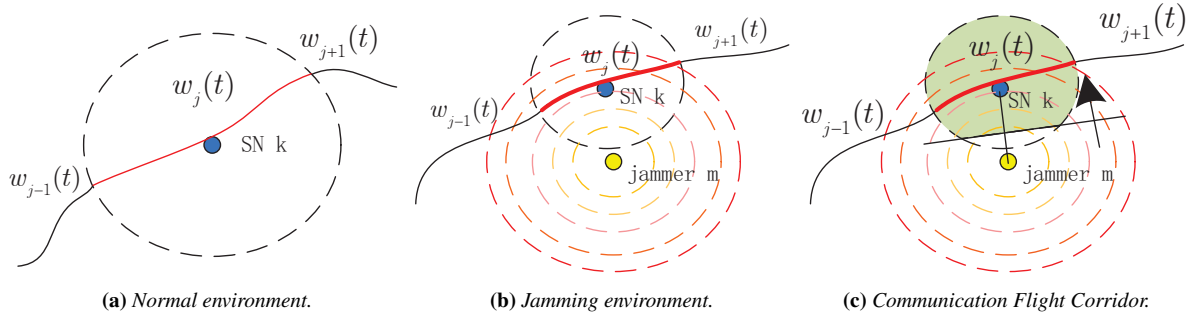


Figure 3. (a) According to the piecewise polynomial locus selected by the initial path, the data collection segment of which is in a circular region. (b) The piecewise polynomial trajectory in the jamming environment, in which the data collection segment is no longer in a circle. (c) The piecewise polynomial locus in CFC, which can meet the communication requirements.

consisting of piecewise polynomials is parametrized to the time variable t , which can be represented by

$$w(t) = \begin{cases} w_1(t) = \sum_{i=0}^n p_{1,i} t^i, T_0 \leq t \leq T_1 \\ w_2(t) = \sum_{i=0}^n p_{2,i} t^i, T_1 \leq t \leq T_2 \\ \vdots \\ w_J(t) = \sum_{i=0}^n p_{J,i} t^i, T_{J-1} \leq t \leq T_J \end{cases}, \quad (13)$$

where $T_{j-1} \leq t \leq T_j$, $j \geq 1$ represents the flight time of j -th trajectory.

The trajectories are divided into two categories. One is outside the SN communication region, and the other is inside. As shown in Figure 3a, the dotted circle is the effective communication area of SN k . Data collection can only be carried out when the UAV is in the effective communication region, which can meet $R_k(t) \geq R_{\min}$, as shown in the red line. We assume that the UAV collects SN data at a fixed transmission rate $R_{\min}$. Therefore, when adopting piecewise trajectory optimization, we need to satisfy $T_j \geq \frac{Q_{\min}}{R_{\min}}$ to complete the mission of data collection. However, as shown in Figure 3b, in a jamming environment, the effective communication range of SN k is no longer a “circle”, leading to the need to find an area that can meet the communication rate requirements when we carry out trajectory planning. The constraint (11b) in problem P1 is difficult to deal with because it is a non-convex constraint and inefficient when used traditional approximation algorithms such as SCA. For

this reason, we proposed a concept of CFC, which can meet the communication conditions of UAV as the green area shown in Figure 3c. Therefore, in trajectory optimization, we only need to constrain the trajectory segment of data collection in CFC to satisfy $R_k(t) \geq R_{\min}$.

In order to find CFC, We define the effective communication region of SN k without jammer as $C_{k,j}(w_k) = \{w_j(t) | (w_j(t) - w_k)^T (w_j(t) - w_k) \leq d_{\max}^2\}$, where $w_j(t)$ represents the trajectory in the region of SN k effective communication and $d_{\max}$ is the maximum distance. To generate the convex CFC from SN and jammers, the procedures are as follows:

1. *Find the nearest jammer:* Calculating the distance between SN k and all the jammers, select the nearest jammer m^* and set the jamming value of the other jammers as $I_{m,k} = \min\{\beta_0 p_m d_{k,m}^{-\alpha}, \beta_0 p_m d_{\max}^{-\alpha}\}$, which is a worst-case scenario that we consider, because the jamming value of the interferer m wouldn't exceed $I_{m,k}$ in the communication region of SN k .
2. *Find critical points:* Searching for a point $P_{m^*,k}$ in the direction of the centre of the SN communication circle on the line segment between the central position of SN k and jammer m^* , which can satisfy the communication rate greater than $R_{\min}$.
3. *Find CFC:* going through $P_{m^*,k}$ and doing the perpendicular l_{k,m^*} on the line between SN k and jammer m^* .

The intersection of the half-space on the side of vertical l_{k,m^*} of SN k and the communication range of SN

k is the CFC in the jamming environment, which is a convex polyhedron. The specific construction process is shown in **Algorithm 1**.

IV. TRAJECTORY OPTIMIZATION

4.1 Piecewise Trajectory Design

In trajectory optimization, we need to optimize the trajectory segment in free space and the trajectory segment in CFC, respectively. We use the Bernstein polynomial basis similar to [32] and represent the trajectory as piecewise Bézier curve, which has a lower complexity for trajectory generation than traditional monomial polynomial one. The polynomial consists of Bernstein basis is called a Bézier curve, and is written as:

$$B_j(t) = c_j^0 b_n^0(t) + c_j^1 b_n^1(t) + \cdots + c_j^n b_n^n(t) = \sum_{i=0}^n c_j^i b_n^i(t), \quad (14)$$

where $[c_j^1, c_j^2, \dots, c_j^n]^\dagger$ denotes as c_j is the set of control points of the j -th piece of the Bézier curve. The Bernstein polynomial basis is defined as $b_n^i(t) = \binom{n}{i} \cdot t^i \cdot (1-t)^{n-i}$, where n is the degree of the basis. Compared to monomial basis polynomial, Bézier curve has several special properties [31, 32]:

1. Endpoint interpolation property. The Bézier curve always starts at the first control point, ends at the last control point, and never pass any other control points.
2. Fixed interval property. The Bézier curve parametrized to variable t is defined on $t \in [0, 1]$.
3. Convex hull property. The Bézier curve $B(t)$ consists of a set of control points c_i are entirely confined within the convex hull defined by all these control points.
4. Hodograph property. The derivative curve $B'(t)$ of a Bézier curve $B(t)$ is called as hodgraph, and it is still Bézier curve with the controls points defined by $c_i^{(1)} = n \cdot (c_{i-1} - c_i)$, where n is the degree.

Actually, for a Bézier curve, the control points can be viewed as the weights of the basis and also the basis can be viewed as the weights of control points. The

piecewise trajectory expressed by the Bézier curve is defined as follows by adopting the method in [32]:

$$f(t) = \begin{cases} s_1 \cdot \sum_{i=0}^n c_1^i b_n^i(\frac{t-T_0}{s_1}), T_0 \leq t \leq T_1 \\ s_2 \cdot \sum_{i=0}^n c_2^i b_n^i(\frac{t-T_1}{s_2}), T_1 \leq t \leq T_2 \\ \vdots \\ s_J \cdot \sum_{i=0}^n c_J^i b_n^i(\frac{t-T_{J-1}}{s_J}), T_{J-1} \leq t \leq T_J \end{cases}, \quad (15)$$

where c_j^i is the i -th control point of the j -th segment of the trajectory. $T_1, T_2, \dots, T_m$ are the end times of each segment. The total time is $T = T_m - T_0$. $s_1, s_2, \dots, s_m$ are the scale factors applied on each piece of the Bézier curve, to scale up the time interval form $[0, 1]$ to time $[T_{i-1}, T_i]$ allocated in one segment.

Accordingly, the j -th segment of the trajectory optimization goal in $P1$ can be rewritten as

$$J_j = \int_{T_{j-1}}^{T_j} \left(\frac{d^k(s_j \cdot \sum_{i=0}^n c_j^i b_n^i(\frac{t}{s_j}))}{dt^k} \right)^2 dt, \quad t \in [T_{j-1}, T_j], \quad (16)$$

Substitute $\tau = s_j t$ into formula (16):

$$\begin{aligned} J_j &= \int_0^1 s_j^3 \left(\frac{d^k(\sum_{i=0}^n c_j^i b_n^i(\tau))}{s_j^k d\tau^k} \right)^2 d\tau, \quad \tau \in [0, 1] \\ &= s_j^{3-2k} \int_0^1 \left(\frac{d^k(\sum_{i=0}^n p_i \tau^i)}{d\tau^k} \right)^2 d\tau \\ &= s_j^{3-2k} \int_0^1 (w^{(k)}(\tau))^2 d\tau \\ &= \mathbf{p}_j^\dagger \cdot \mathbf{Q}_j \cdot \mathbf{p}_j \end{aligned}, \quad (17)$$

where $\mathbf{p}_j = [p_j^1, p_j^2, \dots, p_j^n]^\dagger$ is the j -th piece of polynomial coefficient in formula (13) and existing a matrix $\mathbf{M}$ that can make $\mathbf{p}_j = \mathbf{M} \cdot c_j$ [32]. After the order and allocation time of the polynomial are determined, $\mathbf{Q}_j$ is a constant. When $k = 4$, it can be express

$$\mathbf{Q}_j = s_j^{-5} \begin{bmatrix} & & & \vdots & & \\ \cdots & \frac{i(i-1)(i-2)(i-3)l(l-1)(l-2)(l-3)}{i+l-7} & \cdots & & & \\ & & & \vdots & & \end{bmatrix}. \quad (18)$$

where $i = \{1, 2, \dots\}$ and $l = \{1, 2, \dots\}$ represents

Algorithm 1. Given the position of SN k and the position of jammer, find CFC _{k} .

```

1: function Find_CFC( $w_k, w_j(t), w_m$ )
2: Set the jammer set  $jammer\_open\_list = \{1, 2, \dots, M\}$  and  $jammer\_close\_list = \{\emptyset\}$ .
3: Search for jammer closest to the sensor node:
4:  $m^* = \arg \min_{m \in \mathcal{M}} \|w_k - w_m\|$ 
5: Remove  $m^*$  from the set:
6:  $jammer\_open\_list \rightarrow m^*$ 
7:  $jammer\_close\_list \leftarrow m^*$ 
8: Set the energy of the remaining jammer:
9:  $I_{m,k} = \min\{\beta_0 p_m d_{k,m}^{-\alpha}, \beta_0 p_m d_{\max}^{-\alpha}\}$ 
10: Get the line segment  $[w_k, w_{m^*}]$  between the nearest jammer, and search for point  $P_{m^*,k}$  to make it meet:
    
$$P_{m^*,k} = \arg \max_{P_{m,k} \in [w_k, w_{m^*}]} \|P_{m,k} - w_k\|$$

11:  $s.t. \quad B \cdot \log_2 \left( 1 + \frac{\beta_0 p_k \|w_j(t) - w_k\|^{-\alpha}}{I_k + \sigma^2} \right) \geq R_{\min}$ 
    
$$I_k = \sum_{m=0, m \neq m^*}^M I_{m,k} + \beta_0 p_k \|P_{m,k} - w_k\|^{-\alpha}$$

12: if there is no  $P_{m^*,k}$ , it is determined that SN  $k$  cannot perform the data collection, stop algorithm.
13: else continue
14: Find the perpendicular of line segment  $[w_k, w_{m^*}]$  through point  $P_{m^*,k}$ :
15:  $a_{m^*}^T w = b_{m^*}$ 
16: Search for jammer  $m'$  belonging to  $C_{k,j}(w_k)$  and perpendicular to line segment  $[w_k, w_{m^*}]$  through  $w_{m'}$ :
17:  $a_{m'}^T w = b_{m'}$ 
18: combine perpendiculars:
19:  $\mathbf{A}_k^\dagger \leftarrow \begin{bmatrix} a_0^T \\ a_1^T \\ \vdots \end{bmatrix}, \mathbf{b}_k \leftarrow \begin{bmatrix} b_0 \\ b_1 \\ \vdots \end{bmatrix}$ 
20: Calculate the CFC:
21:  $\Omega_{k,j} = C_{k,j}(w_k) \cap \{\mathbf{A}_k^\dagger \cdot w_j(t) \leq \mathbf{b}_k\}, \Omega_j \leftarrow \Omega_{k,j}$ 
22: end

```

the coefficient label of the polynomial. So we only need to take $\mathbf{c}_j$ as the optimization variable. Meanwhile, due to the nature of Bézier curve, the complexity of the optimization problem with $\mathbf{c}_j$ as the optimization variable is much lower than that of the ordinary polynomial coefficient $\mathbf{p}_j$. According to CFC and the UAV trajectory represented by the piecewise

Bézier curve determined by the initial path, our trajectory optimization problem $P1$ for UAV data collection can be rewritten as

$$(P3) \quad \min_{\mathbf{c}} \sum_{j=1} \mathbf{c}_j^\dagger \mathbf{Q}'_j \mathbf{c}_j, \quad (19a)$$

$$s.t. \quad \mathbf{A}_{eq} \mathbf{c} = \mathbf{b}_{eq}, \quad (19b)$$

$$\mathbf{A}_{ieq} \mathbf{c} \leq \mathbf{b}_{ieq}, \quad (19c)$$

$$\mathbf{c}_j \in \Omega_j, j = 1, 2, \dots, J, \quad (19d)$$

where $P3$ is a standard QCQP and $\mathbf{Q}'_j = \mathbf{M}^\dagger \mathbf{Q}_j \mathbf{M}$, since the optimization objective $\sum_{j=1} \mathbf{c}_j^\dagger \mathbf{Q}'_j \mathbf{c}_j$ is quadratic function, equality constraint (19b) and inequality constraint (19c) are affine functions, and Ω_j in constraint (19d) is the intersection of quadratic and affine functions. Meanwhile, equality constraint (19b) mainly guarantees two aspects: 1) Boundary-point constraint: According to property 1, we only need to ensure that the first control point is equal to the starting point position and the last control point is the end point position. $c_1^0 = w_0, c_J^n = w_T$. 2) Continuity constraint: we ensure that the control point in the previous curve is equal to the first control point in the next curve: $c_j^{(k),n} \cdot s_j^{1-k} = c_{j+1}^{(k),0} \cdot s_{j+1}^{1-k}$ between each piecewise Bézier curve. Inequality constraint (19c) restricts the velocity $v_{\min} \leq n \cdot (c_j^i - c_j^{i-1}) \leq v_{\max}$, and acceleration $a_{\min} \leq n \cdot (n-1) \cdot (c_j^i - 2c_j^{i-1} + c_j^{i-2}) / s_j \leq a_{\max}$ of the UAV within a certain range which could be obtained from property 4. According to property 3, constraint in (19d) $\Omega_j = \bigcup_k \Omega_{k,j}$ is a reliable data collection area which is computed in **Algorithm 1**.

4.2 Time Allocation

The time allocation significantly affects the resulting trajectories. We divide the trajectory of UAV into two categories, namely the trajectory in the free space $w_j(t), j \in \mathcal{J}_{CFC}$ and the trajectory in the CFC $w_j(t), j \in \mathcal{J}_{free}$, where $\mathcal{J}_{CFC} \cup \mathcal{J}_{free} \subseteq \mathcal{J} = \{1, 2, \dots, J\}$. In the past researches on obstacle avoidance or safe flight of UAV, commonly used methods for time allocation include trapezoidal method allocation, average allocation, distance proportion allocation, etc.[31, 32]. For two different strategies, hover mode and continuous flight mode, we set the total flight time T of the UAV based on trapezoidal method

allocation considering the demand of data collection amount $Q_{\min}$.

In hover mode, since the velocity and acceleration at each hover point are 0, the trapezoidal calculation principle should be adopted for each piecewise trajectory, as shown below

$$T_h = T_{f,h}^{\min} + T_{collect}^{\min}, \quad (20)$$

where $T_{f,h}^{\min} = \sum_{j \in \mathcal{J}} (\frac{d_j}{\sqrt{2}v_{\max}} + \frac{v_{\max}}{a_{\max}})$ and $T_{collect}^{\min} = \frac{Q_{\min}}{R_{\min}}$. d_j is the Euclidean distance between two way-points on the trajectory of segment j -th on the initial path. T_h is the total mission time required by the hover mode when the amount of data collected is $Q_{\min}$.

In continuous flight mode, we only need to calculate the time for the whole segment, which is expressed as follows

$$T_c = T_{f,c}^{\min} + \sum_{j \in \mathcal{J}_{CFC}} \max\{T_{collect}^{\min} - \Delta T_j, 0\}. \quad (21)$$

where $T_{f,c}^{\min} = \sum_{j \in \mathcal{J}} \frac{d_j}{\sqrt{2}v_{\max}} + \frac{v_{\max}}{a_{\max}}$. ΔT_j is the time of the piecewise trajectory at segment j and T_c is the continuous flight time when the amount of data collected is $Q_{\min}$. The calculation process of time allocation of trapezoidal method allocation is given in Appendix B.

4.3 Convergence and Complexity Analysis

The trajectory optimization of UAV in data collection mission mainly consists of three parts, including initial path planning, CFC construction and trajectory optimization. In the initial path planning, the DP is adopted, whose convergence performance can be proved in [38], and the complexity of DP is $O(K^3)$. CFC construction is to obtain the effective communication region through geometric methods, whose complexity is only related to the number of jammers and sensors, so its complexity is $O(KM)$. The trajectory optimization problem in P3 is a standard QCQP, which has $O(3(n+1))$ starting point constraints and $O(3(n+1))$ end point constraints, $O(2(n+1)(J-1))$ state continuity constraints, $O(2(n+1)(J-1))$ velocity range constraints, $O(2(n+1)(J-1))$ acceleration range constraints, and $O((n+1)J_{CFC})$ CFC constraints, where n is the order of the polynomial, J

Table 1. Key simulation parameters.

The Notation	Physical Meaning	Value
p_0	SN transmit power	1000mW
B	channel bandwidth	20kHz
$v_{\min}, v_{\max}$	UAV maximum velocity	$\pm 30\text{m/s}$
$a_{\min}, a_{\max}$	UAV maximum velocity	$\pm 5\text{m/s}$
σ^2	noise power spectral density	-169dBm/Hz
β_0	The path loss factor	-45dB
α	The path loss exponent	4

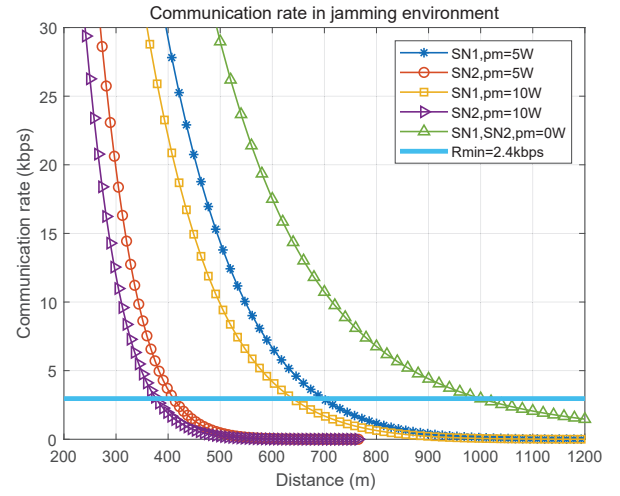


Figure 4. Data collection rate under jamming environments.

is the total number of segments of the trajectory, and J_{CFC} is the number of CFCs.

V. NUMERICAL RESULTS

5.1 Simulation Environments

This subsection provides simulation results to verify the performance of our proposed trajectory optimization algorithm, as compared to the following two strategies: 1) data is collected while hovering over each sensor; 2) data is collected during continuous flight within the communication region of each sensor as perfect information. Some key parameters setting are listed in Table 1.

In the simulation, the positions of two SNs are set as (1200, 2400) and (3600, 3000), and the positions of the two jammers are (1300, 1000) and (3000, 2500), which are in meters. We set the transmission power of each SN and jammer to be 1W, and we can calculate the communication rate change graph of SN

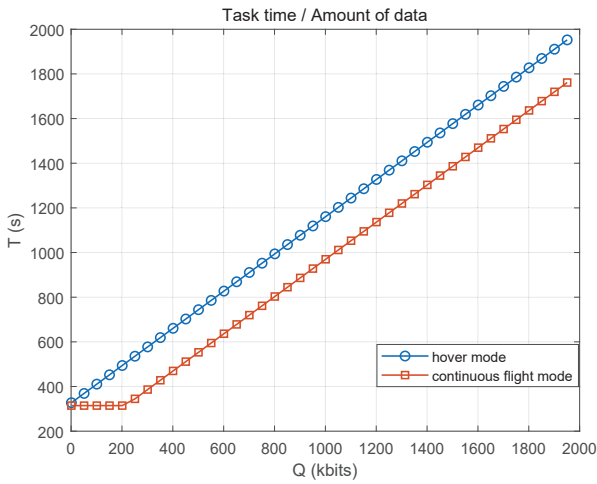


Figure 5. Diagram of data collection and minimum mission time.

and the connection nearest to the jammer according to the method in **Algorithm 1**. As shown in Figure 4, the horizontal axis is the distance between the UAV and SN, and the vertical axis is the instantaneous communication rate obtained by the UAV. We set the rate of the communication system in this scene as $R_{\min} = 2.4k\text{bps}$. According to the parameter settings in table 1, we can calculate that the rate obtained by the UAV within 1km from the SN without jamming can meet the communication requirements. As shown in Figure 4, when the jammer exists, the effective transmission range of the SN is no longer a 'circle' with a radius of 1km , so we plan the effective transmission region of SN.

5.2 Simulation Results

The initial path planning of data collection is a TSP problem, which has many mature solutions. In the scenario of this paper, the initial path can directly obtain the line segment, which is the starting point to SN 1 to SN 2 to end point. We select the waypoints (50, 50), (678,1546), (2158,2687), (2622,2789), (4448,2469), (5000, 2000) to allocate the time of the piecewise trajectory based on the initial path in continuous flight strategy. But in the hover strategy, we directly use the starting point and finishing point of the drone and the position of the SN as waypoints. The mission time under different data collection requirements was investigated, as shown in Figure 5. The blue line in Figure 5 is always above the red line, indicating that hovering mode consistently takes more mission time than

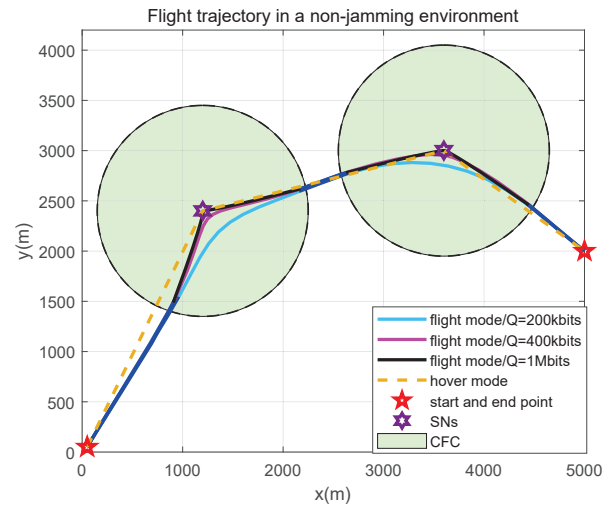


Figure 6. Flight trajectory in a non-jamming environment.

continuous flight mode, which is consistent with our general reasoning. Meanwhile, for continuous flight mode, when the amount of data collected is less than 200kbits , the flight time is fixed at about 314.5s ; when the amount of data collected is more significant than 200kbits , the flight time has a linear relationship with the amount of data collected. The hover mode is always proportional to the amount of data collected.

Figure 6 shows the flight trajectories of hover mode and continuous flight mode in a non-jamming environment with different data amount requirements. It can be seen that no matter what communication data demand is, the flight path of hover mode is fixed as the connection between each waypoint. However, the trajectory of continuous flight mode under different data demand shows a trend that the larger the data amount is, the closer the flight path is to the waypoint. It should be noted that in the hover mode, the UAV flies along a fixed way regardless of the presence of jamming.

Figure 7 shows the velocity changes of the UAV under different parameters. The hover mode's velocity accelerates from 0 to the maximum and then decelerates to 0 between any two waypoints under any data acquisition requirements. The flight mode also accelerates to the target waypoint and then decelerates during the flight. However, the more significant the amount of data collected, the longer the flight takes, the more dramatic the change in velocity will be. Figure 8 shows the flight trajectory of the UAV under $p_m = 10\text{W}$ in the jamming environment. It can be

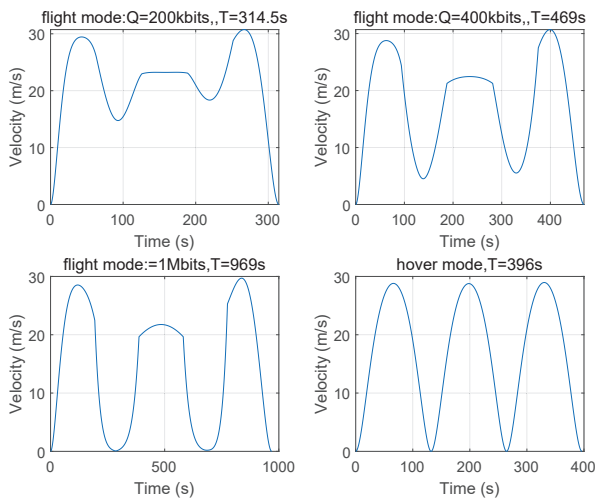


Figure 7. Flight velocity in a non-jamming environment.

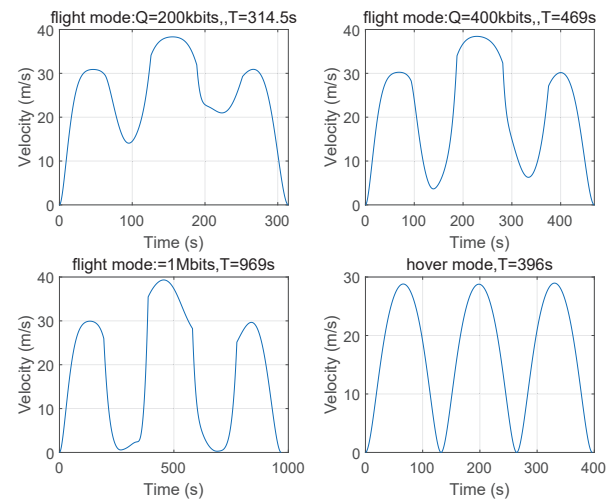


Figure 9. Flight velocity in jamming environment where $p_m = 10W$.

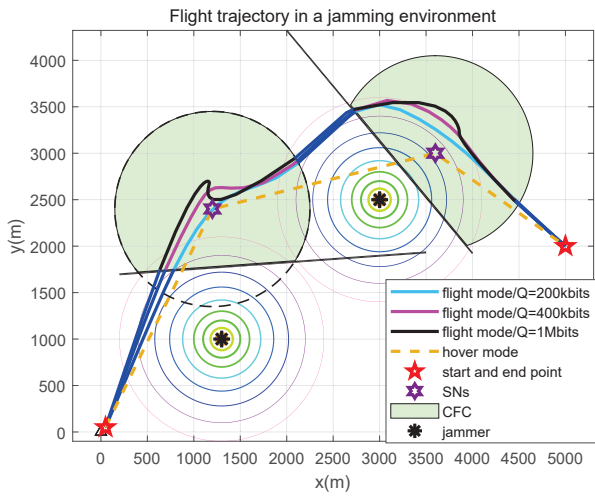


Figure 8. Flight trajectory in jamming environment where $p_m = 10W$.

intuitively seen that due to the presence of jammers, the range of CFC is cut into a 'semicircle' from the 'circle' in Figure 6, and effective data collection will be completed in the green 'semicircle'. Figure 9 shows the change curve of the UAV's flight velocity in a disturbed environment. Compared with the change curve of the UAV's flight velocity without interferers in Figure 7, it can be seen that the change of UAV's flight velocity in a disturbed environment is more distant, which means it will consume more energy. Figure 10 shows the energy consumption of UAVs configured with different mission times without considering the amount of data collected. The energy consumption here adopts the optimized target value in P1, which has a consistent changing relationship with the

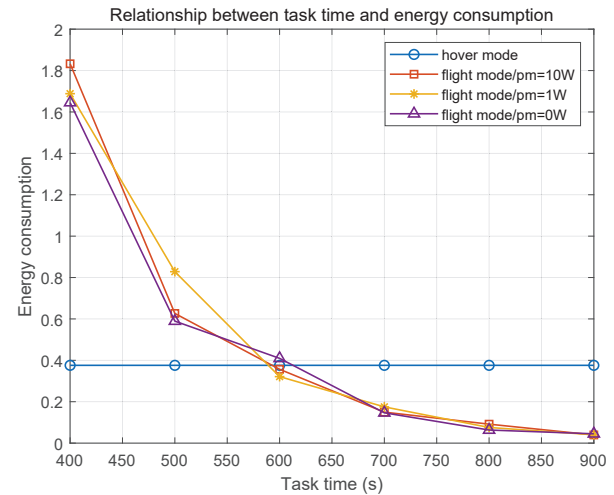


Figure 10. Relationship between mission time and energy consumption.

actual energy consumption. When the mission time is longer than 600s, the energy consumption of continuous flight mode is less than that of hover mode. Since the hover mode also consumes energy during the hover process, the total energy consumed in the hover mode is not necessarily occupied even when the mission time is less than 600s. Although the flight energy consumption of UAV is similar under several simulated powers, in some specific task time conditions, the difference of time allocated for each trajectory under fixed time allocation scheme affects the flight trajectory and energy consumption of UAV, so some crossover points are generated. It can be seen that the flight trajectory and energy consumption of

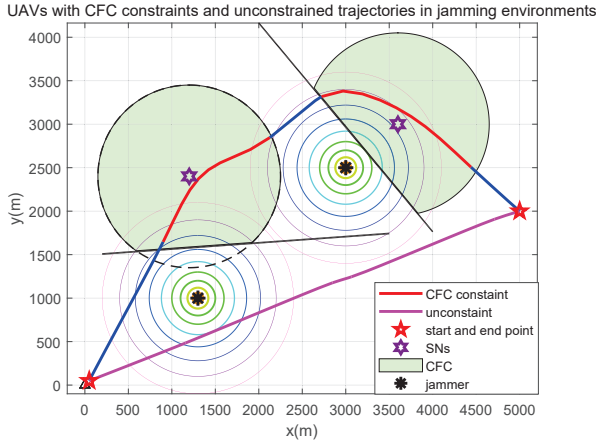


Figure 11. UAV trajectory without CFC constraints and UAV trajectory under CFC constraints where $Q=200\text{kbits}$, $p_m=5W$, $T=314.5\text{s}$.

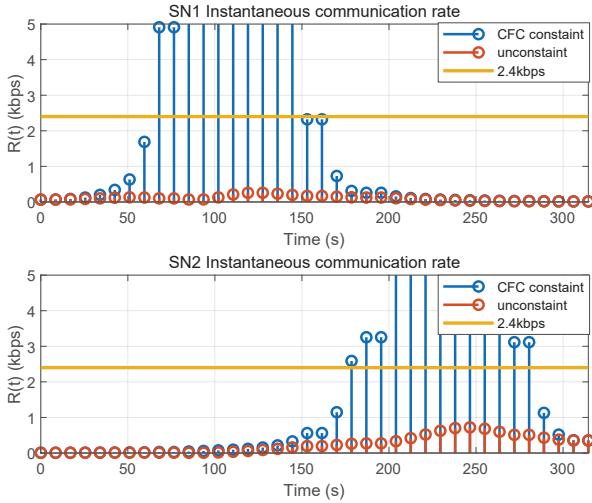


Figure 12. Instantaneous communication rate.

UAV are sensitive to the assigned task time, and the optimal time allocation scheme will be the direction of our next research.

Figure 11 shows the flight trajectory with and without CFC constraints under $p_m = 5W$. It can be seen that the flight trajectory of the UAV without CFC constraints is not completed within the effective communication area, which is the connection from the starting point to the end point. Figure 12 shows the instantaneous communication rate of UAV with and without CFC constraints. It can be seen that when there is no constraint, the communication rate of UAV always fails to meet the minimum communication requirements.

VI. CONCLUSION

In this paper, we study the trajectory optimization of UAV data collection mission in a jamming environment. Aiming at the problems existing in traditional SCA and heuristic algorithms, namely initial sensitivity and discrete accuracy, the concept of communication corridor is proposed for the first time, and the standard convex optimization problem is constructed by piecewise B ezier curve. Simulation results validate the effectiveness of the proposed algorithm.

This paper provides several insights in understanding the trajectory optimization problem. From the communication perspective, the UAV's behaviours are of paramount importance and need to be carefully designed, including their locations, velocity, and accelerations. The UAV flight procedure can be divided into three stages: i) Initial path planning; ii) CFC construct; iii) Trajectory optimization and time allocation. Therefore, in practice, we can independently design the UAV paths at each stage to reduce the computational complexity. Data collection can only occur when the communication link between the UAV and the sensor meets certain conditions. Therefore, sensors are not required to continuously transmit information signals in the whole procedure to improve energy efficiency. In future work, We consider the communication corridor design and UAV trajectory optimization in the presence of active jammers. At the same time, in addition to minimizing energy as the optimization goal, we also further consider the optimization of mission time.

APPENDIX A

After obtaining the adjacency matrix, we first calculate the dynamic programming value of the corresponding V' subset of each optional node according to

$$DP(i, V') = \min_{V'} \{d_{i,k} + DP(k, V' - \{k\})\}. \quad (22)$$

The binary representation of a subset is defined. The binary representation of each number is $(1 \ll (k))$, and k is the number. If $k = 1$, it is an empty set. If there are several node numbers in the subset, then the values of all numbers are added up to the index of the final DP. We start searching for the optimal path once the DP table is obtained. $DP(1, \{2, 3, 4, \dots, K+1\})$

Algorithm 2. *Dynamic programming algorithm for (P2).*

```
1: Initilization: Input waypoint number  $K$  and adjacency matrix  $cost\_matrix$ , index of SN numbered subsets  $2^K$ , define value table  $dp(K+1,K)$ .
2: Compoute DP value table:
3: Loop:  $k = 1, \dots, K$ 
4:    $dp(k, 1) = cost\_matrix(n, K + 1)$ 
5: End Loop
6: Loop:  $m = 1, \dots, 2^K$ 
7:   Loop:  $k = 1, \dots, K$ 
8:      $dp(k, m) = \inf$ 
9:     if  $k$  is not in the subset  $m$ 
10:      continue
11:     Loop:  $n = 1, \dots, K + 1$ 
12:       if  $n$  is in the subset  $m$ 
13:         continue
14:         if  $dp(k, m) > c(k, n) + dp(k, (1 \ll (n - 1)) \oplus m)$ 
15:            $dp(k, m) = c(k, n) + dp(k, (1 \ll (n - 1)) \oplus m)$ 
16:       End Loop
17:   End Loop
18: End Loop
19: Search optimal path:
20: Define PATH_INDEX and put in the starting position number, set path pointer=1, min = inf
21: Repeat
22:   Loop:  $k = 2, \dots, K$ 
23:     if  $k$  is in the subset  $V'$  & min  $> cost\_matrix(k, pointer) + dp(2^K, k - 1)$ 
24:       min =  $cost\_matrix(k, pointer) + dp(2^K, k - 1)$ 
25:     temp =  $k$ 
26:   End Loop
27:   remove  $k$  form  $V'$  and store PATH_INDEX
28: Until:  $V' = \emptyset$ 
```

is regarded to keep searching until it's empty. **Algorithm 2** gives the process of solving TSP by dynamic programming.

APPENDIX B

Time allocation for each segment of data is conducted according to the trapezoidal principle. Set acceleration time t_1 , deceleration time t_2 , full velocity flight time

t_3 , and flight distance d , which can be obtained

$$d = \frac{1}{2}\sqrt{2}a_{\max}t_1^2 + \sqrt{2}v_{\max}t_3 + \frac{1}{2}\sqrt{2}a_{\max}t_2^2. \quad (23)$$

where $v_{\max} = a_{\max}t_1 = a_{\max}t_2$. Substitute $t_1 = t_2 = \frac{v_{\max}}{a_{\max}}$ into Equation (16)

$$t_3 = \frac{d - \sqrt{2}\frac{v_{\max}^2}{a_{\max}}}{\sqrt{2}v_{\max}} = \frac{d}{\sqrt{2}v_{\max}} - \frac{v_{\max}}{a_{\max}}, \quad (24)$$

So the total flight time required is

$$T = t_1 + t_2 + t_3 = \frac{d}{\sqrt{2}v_{\max}} + \frac{v_{\max}}{a_{\max}}. \quad (25)$$

References

- [1] G. Ding, Q. Wu, *et al.*, "An amateur drone surveillance system based on the cognitive internet of things," *IEEE Communications Magazine*, vol. 56, no. 1, 2018, pp. 29–35.
- [2] X. Zhang, D. Guo, *et al.*, "Auction-based multichannel cooperative spectrum sharing in hybrid satellite-terrestrial iot networks," *IEEE Internet of Things Journal*, vol. 8, no. 8, 2021, pp. 7009–7023.
- [3] W. Wang, N. Cheng, *et al.*, "Content delivery analysis in cellular networks with aerial caching and mmwave backhaul," *IEEE Transactions on Vehicular Technology*, vol. 70, no. 5, 2021, pp. 4809–4822.
- [4] S. Hu, W. Ni, *et al.*, "Joint optimization of trajectory, propulsion, and thrust powers for covert uav-on-uav video tracking and surveillance," *IEEE Transactions on Information Forensics and Security*, vol. 16, 2021, pp. 1959–1972.
- [5] J. Tang, G. Chen, *et al.*, "Secrecy performance analysis of wireless communications in the presence of uav jammer and randomly located uav eavesdroppers," *IEEE Transactions on Information Forensics and Security*, vol. 14, no. 11, 2019, pp. 3026–3041.
- [6] Y. Wang, L. Chen, *et al.*, "Resource allocation and trajectory design in uav-assisted jamming wideband cognitive radio networks," *IEEE Transactions on Cognitive Communications and Networking*, vol. 7, no. 2, 2021, pp. 635–647.
- [7] A. Rahmati, S. Hosseinalipour, *et al.*, "Lifetime maximization for uav-assisted data gathering networks in the presence of jamming," in *2020 IEEE 21st International Workshop on Signal Processing Advances in Wireless Communications (SPAWC)*, 2020, pp. 1–5.
- [8] W. Huang, Z. Yang, *et al.*, "Joint power, altitude, location and bandwidth optimization for uav with underlaid d2d communications," *IEEE Wireless Communications Letters*, vol. 8, no. 2, 2019, pp. 524–527.
- [9] M. B. Ghorbel, D. Rodríguez-Duarte, *et al.*, "Joint position and travel path optimization for energy efficient wireless data gathering using unmanned aerial vehicles," *IEEE*

- Transactions on Vehicular Technology*, vol. 68, no. 3, 2019, pp. 2165–2175.
- [10] J. Zong, C. Shen, *et al.*, “Flight time minimization via uav’s trajectory design for ground sensor data collection,” in *2019 16th International Symposium on Wireless Communication Systems (ISWCS)*, 2019, pp. 255–259.
 - [11] C. Zhan and Y. Zeng, “Completion time minimization for multi-uav-enabled data collection,” *IEEE Transactions on Wireless Communications*, vol. 18, no. 10, 2019, pp. 4859–4872.
 - [12] H. Wang, J. Wang, *et al.*, “Completion time minimization with path planning for fixed-wing uav communications,” *IEEE Transactions on Wireless Communications*, vol. 18, no. 7, 2019, pp. 3485–3499.
 - [13] Y. Wu, W. Yang, *et al.*, “Energy-efficient trajectory design for uav-enabled communication under malicious jamming,” *IEEE Wireless Communications Letters*, vol. 10, no. 2, 2021, pp. 206–210.
 - [14] C. Zhan, Y. Zeng, *et al.*, “Energy-efficient data collection in uav enabled wireless sensor network,” *IEEE Wireless Communications Letters*, vol. 7, no. 3, 2018, pp. 328–331.
 - [15] Y. Zeng, J. Xu, *et al.*, “Energy minimization for wireless communication with rotary-wing uav,” *IEEE Transactions on Wireless Communications*, vol. 18, no. 4, 2019, pp. 2329–2345.
 - [16] Y. Wang, X. Wen, *et al.*, “Multi-uav collaborative data collection for iot devices powered by battery,” in *2020 IEEE Wireless Communications and Networking Conference (WCNC)*, 2020, pp. 1–6.
 - [17] F. Zhou, Y. Wu, *et al.*, “Computation rate maximization in uav-enabled wireless-powered mobile-edge computing systems,” *IEEE Journal on Selected Areas in Communications*, vol. 36, no. 9, 2018, pp. 1927–1941.
 - [18] T. M. Nguyen, A. Yadav, *et al.*, “Centralized and distributed energy efficiency designs in wireless backhaul hetnets,” *IEEE Transactions on Wireless Communications*, vol. 16, no. 7, 2017, pp. 4711–4726.
 - [19] M. Samir, S. Sharafeddine, *et al.*, “Uav trajectory planning for data collection from time-constrained iot devices,” *IEEE Transactions on Wireless Communications*, vol. 19, no. 1, 2020, pp. 34–46.
 - [20] C. Chang, S. W. Loke, *et al.*, “An energy-efficient inter-organizational wireless sensor data collection framework,” in *2015 IEEE International Conference on Web Services*, 2015, pp. 639–646.
 - [21] Y. Zeng, X. Xu, *et al.*, “Trajectory design for completion time minimization in uav-enabled multicasting,” *IEEE Transactions on Wireless Communications*, vol. 17, no. 4, 2018, pp. 2233–2246.
 - [22] J. Peng, Z. Zhang, *et al.*, “Anti-jamming communications in uav swarms: A reinforcement learning approach,” *IEEE Access*, vol. 7, 2019, pp. 180 532–180 543.
 - [23] Y. Xu, G. Ren, *et al.*, “A one-leader multi-follower bayesian-stackelberg game for anti-jamming transmission in uav communication networks,” *IEEE Access*, vol. 6, 2018, pp. 21 697–21 709.
 - [24] Y. Xu, G. Ren, *et al.*, “Joint power and trajectory optimization in uav anti-jamming communication networks,” in *ICC 2019 - 2019 IEEE International Conference on Communications (ICC)*, 2019, pp. 1–5.
 - [25] B. Duo, Q. Wu, *et al.*, “Anti-jamming 3d trajectory design for uav-enabled wireless sensor networks under probabilistic los channel,” *IEEE Transactions on Vehicular Technology*, vol. 69, no. 12, 2020, pp. 16 288–16 293.
 - [26] H. Wang, J. Chen, *et al.*, “Trajectory planning in uav communication with jamming,” in *2018 10th International Conference on Wireless Communications and Signal Processing (WCSP)*, 2018, pp. 1–6.
 - [27] N. Gao, Z. Qin, *et al.*, “Anti-intelligent uav jamming strategy via deep q-networks,” *IEEE Transactions on Communications*, vol. 68, no. 1, 2020, pp. 569–581.
 - [28] H. Jiang, Z. Zhang, *et al.*, “Three-dimensional non-stationary wideband geometry-based uav channel model for a2g communication environments,” *IEEE Access*, vol. 7, 2019, pp. 26 116–26 122.
 - [29] H. Jiang, Z. Zhang, *et al.*, “Three-dimensional geometry-based uav-mimo channel modeling for a2g communication environments,” *IEEE Communications Letters*, vol. 22, no. 7, 2018, pp. 1438–1441.
 - [30] H. Jiang, Z. Zhang, *et al.*, “A novel 3d uav channel model for a2g communication environments using aod and aoa estimation algorithms,” *IEEE Transactions on Communications*, vol. 68, no. 11, 2020, pp. 7232–7246.
 - [31] Z. Wang, X. Zhou, *et al.*, “Alternating minimization based trajectory generation for quadrotor aggressive flight,” *IEEE Robotics and Automation Letters*, vol. 5, no. 3, 2020, pp. 4836–4843.
 - [32] S. Liu, M. Watterson, *et al.*, “Planning dynamically feasible trajectories for quadrotors using safe flight corridors in 3-d complex environments,” *IEEE Robotics and Automation Letters*, vol. 2, no. 3, 2017, pp. 1688–1695.
 - [33] S. Liu, K. Mohta, *et al.*, “Search-based motion planning for aggressive flight in se(3),” *IEEE Robotics and Automation Letters*, vol. 3, no. 3, 2018, pp. 2439–2446.
 - [34] S. Liu, N. Atanasov, *et al.*, “Search-based motion planning for quadrotors using linear quadratic minimum time control,” in *2017 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2017, pp. 2872–2879.
 - [35] B. Zhou, J. Pan, *et al.*, “Raptor: Robust and perception-aware trajectory replanning for quadrotor fast flight,” *IEEE Transactions on Robotics*, 2021, pp. 1–18.
 - [36] J. Gu, H. Wang, *et al.*, “Energy-constrained completion time minimization in uav-enabled internet of things,” *IEEE Internet of Things Journal*, vol. 7, no. 6, 2020, pp. 5491–5503.
 - [37] Z. Liu, G. Ding, *et al.*, “Cooperative topology sensing of wireless networks with distributed sensors,” *IEEE Transactions on Cognitive Communications and Networking*, vol. 7, no. 2, 2021, pp. 524–540.
 - [38] C. Hong and A. H. Tewfik, “Heuristic reusable dynamic programming: Efficient updates of local sequence alignment,” *IEEE/ACM Transactions on Computational Biology and Bioinformatics*, vol. 6, no. 4, 2009, pp. 570–582.
 - [39] H. Daryanavard and A. Harifi, “Uav path planning for data gathering of iot nodes: Ant colony or simulated annealing optimization,” in *2019 3rd International Conference on Internet of Things and Applications (IoT)*, 2019, pp. 1–4.
 - [40] T. Yang, Y. Hu, *et al.*, “Genetic algorithm based uav trajectory design in wireless power transfer systems,” in *2019 IEEE Wireless Communications and Networking Conference (WCNC)*, 2019, pp. 1–6.

- [41] B. Alidaee, H. Wang, *et al.*, “A note on integer programming formulations of the real-time optimal scheduling and flight path selection of uavs,” *IEEE Transactions on Control Systems Technology*, vol. 17, no. 4, 2009, pp. 839–843.

Biographies



Binbin Wu received his B.S. and M.S. degree in Nanjing University of Information Science & Technology, Nanjing, China, in 2012 and 2015, respectively. He is currently working toward his Ph.D. degree from Army Engineering University. His research interests focus on unmanned aerial vehicle (UAV) communications, Anti-jamming communications, physical layer security, Reinforcement learning, and Deep learning.



Bangning Zhang received the B.S. degree and M.S. degree in Institute of Communications Engineering (ICE), Nanjing, China, in 1984 and 1987, respectively. He is currently a Full Professor and the Head of College of Communications Engineering. He has authored and coauthored more than 80 conference and journal papers and

has been granted over 20 patents in his research areas. He has served as a reviewer for several journals in communication field. His current research interests include communication anti-jamming technologies, microwave technologies, polarization technologies, satellite communications systems, cooperative communications and physical layer security.



Daoxing Guo received the B.S., M.S., and Ph.D. degrees from the College of Communications Engineering, Nanjing, China, in 1995, 1999, and 2002, respectively. He is currently a Full Professor and a Ph.D. Supervisor with the Army Engineering University of PLA. He has authored or coauthored more than 40 conference and journal papers. He has served as a reviewer for several

journals in communication field. His current research interests include satellite communications systems and transmission technologies, communication anti-jamming technologies, and communication anti-interception technologies, including physical layer security. He holds over 20 patents in his research areas.



Hongbin Wang received the B.S. degree in electronic information engineering from Jiangnan University in 2010 and the M.S. degree in signal and information processing from Jiangnan University in 2014. He is currently working toward his Ph.D. degree from Army Engineering University, Nanjing, China. His research

interests focus on wireless communication, satellite communications, communications reconnaissance, anti-interference communication, and electronic countermeasures.



Hao Jiang (Member, IEEE) received the B.S. and M.S. degrees in electrical and information engineering from the Nanjing University of Information Science and Technology, Nanjing, China, in 2012 and 2015, respectively, and the Ph.D. degree from Southeast University, Nanjing, in 2019. From 2017 to 2018, he was a

Visiting Student with the Department of Electrical Engineering, Columbia University, New York City, NY, USA. Since April 2019, he has been a Professor with the School of Information Science and Engineering, Nanjing University of Information Science and Technology. His current research interests are in the general area of wireless channel measurements and modeling, B5G wireless communication networks, signal processing, machine learning, and AI-driven technologies.